Data Science

Lecture 03: Probabilistic Graphical Models

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Where We Left Off

Lecture 02 gave us conditioning and Bayes for a handful of events. Real questions involve dozens of variables at once — and the assumption that makes them affordable says *which variables do not depend on each other*.

Lecture 02
Bayes for two events

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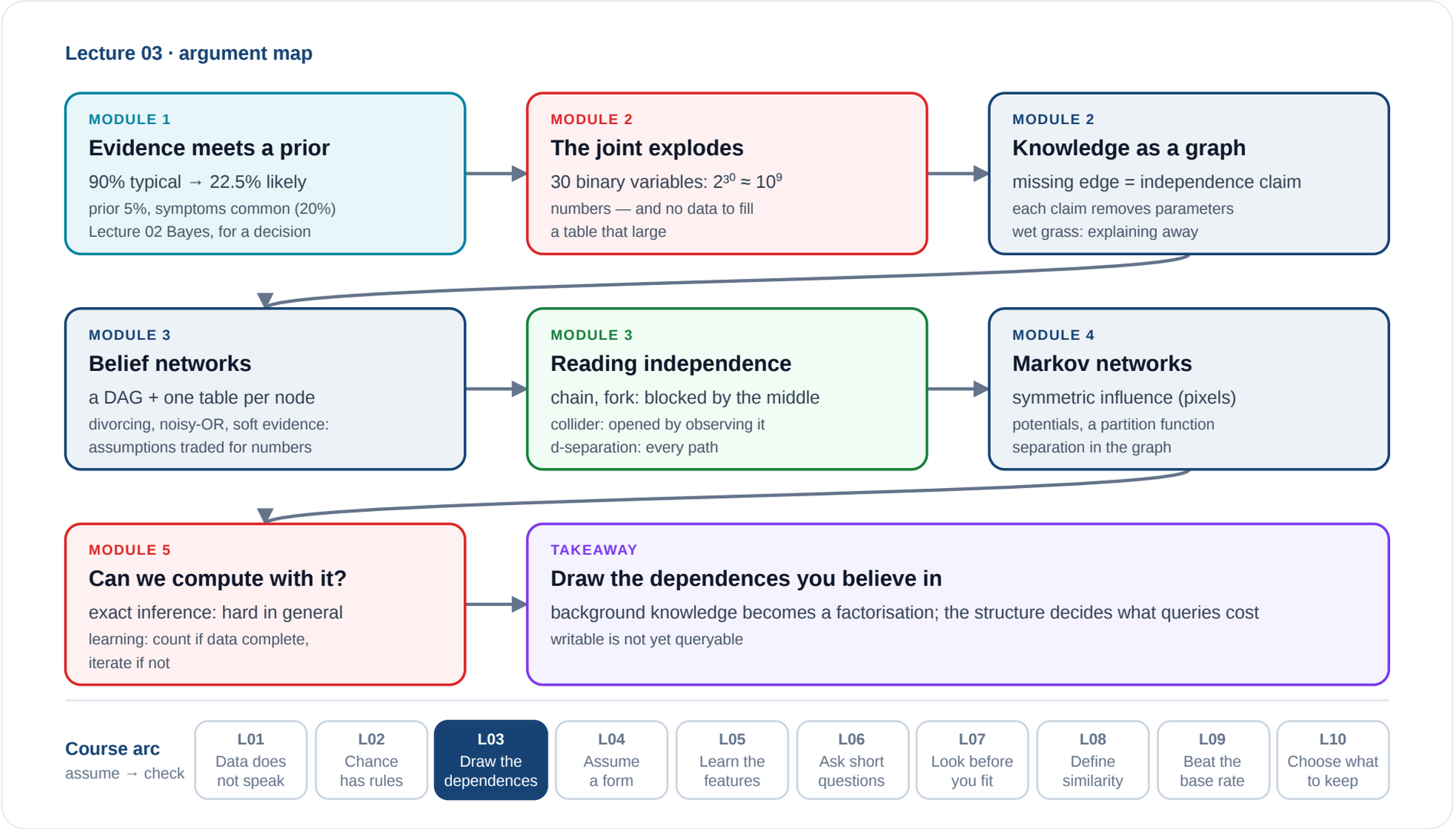
Today
one joint distribution over many variables

→

The assumption
a graph: every missing edge an independence claim

> Lecture 01's churn puzzle — does the app keep customers, or do loyal customers install the app? — is a question about arrows. Module 3 draws both answers.

Argument Map



Module 1

Deciding Under Uncertainty

> ***Guiding Question:*** *If a symptom is 90% typical of a disease, why does having the symptom not make the disease 90% likely?*

Thinking With Uncertainty

Bayes' theorem lets us update a **hypothesis** H in the light of **data** D:

$$P(H \mid D) = \frac{P(H) \cdot P(D \mid H)}{P(D)}$$

This is the whole apparatus of evidence-based decision making in one line: a belief, a measurement of how well the belief explains what we saw, and a normalisation.

Example: Do I Have the Flu?

You have a headache and a cough. From the literature:

Likelihood

90% of people with the flu show headache and coughing

$$P(D \mid H) = 0.9$$

Prior

5% of people currently have the flu

$$P(H) = 0.05$$

Evidence

headache and coughing occur in 20% of people

$$P(D) = 0.2$$

How likely is it that you have the flu?

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How likely is it that you have the flu?

$$P(H \mid D) = \frac{0.05 \cdot 0.9}{0.2} = 0.225$$

Why So Low?

Few people have the flu

The prior $P(H) = 0.05$ caps how far the evidence can carry us.

The symptoms are common

$P(D) = 0.2$ means many people have these symptoms *without* the flu.

A symptom is informative to the degree that it is **rare in general** but **common under the hypothesis**. A likelihood alone tells you nothing.

Takeaway

Bayesian decision making weighs **prior × likelihood** against how ordinary the evidence is. Deciding under uncertainty means keeping all three numbers in view, not just the one the study reports.

Module 2

Why Graphical Models?

> ***Guiding Question:*** *Thirty binary variables need a billion numbers. What knowledge would let us write the same distribution with a few dozen?*

The Problem: Joint Distributions Explode

Consider a joint distribution $p(x_1, \ldots, x_n)$ over n **binary** random variables. How many configurations are there?



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Consider a joint distribution $p(x_1, \ldots, x_n)$ over n **binary** random variables. How many configurations are there?

$$2^n \quad \text{states}, \qquad 2^n - 1 \quad \text{free parameters}$$

Specification

For large n, writing down all values is impossible.

Inference

A single marginal $p(x_1)$ requires summing over 2^{n-1} states.

Chain rule?

It rewrites the joint exactly — but does not reduce the number of values.

What a Graphical Model Is

Definition (Graphical model). A graphical model depicts the dependence and independence relations of a distribution using a graph G as its underlying structure.

Modelling

Describe the variables of a domain and their interactions — how $p(x_1, \ldots, x_n)$ **factorises**.

Inference

Answer queries such as $p(x_1 \mid x_2 = a, \ldots)$ efficiently over that factorisation.

Usually one class of model is used for modelling and another for inference; the latter is often compiled from the former.

How a Graph Can Help

Probabilistic inference is summing and multiplying. Let every variable be a node, and read edges as interaction:

No edge

$$f(a, b) = g(a) \cdot h(b)$$

The variables are independent — the factor splits.

Edge

$$f(a, b) \neq g(a) \cdot h(b)$$

The variables are dependent — the factor does not split.

Edge **direction** can encode further structure. The principle extends to any number of variables.

> Graphical models have limited expressive power: not every conditional independence statement can be drawn, and different graphs can represent the same distribution.

Example: Why Is the Grass Wet?

Tracey leaves her house and sees that her grass is wet. **Was it rain, or did she leave the sprinkler on?** She notices her neighbour Jack's grass is also wet.

R	S	J	T
raining (1) or not (0)	Tracey's sprinkler on (1) or off (0)	Jack's grass wet (1) or dry (0)	Tracey's grass wet (1) or dry (0)

Building the Model

The full joint over four binary variables needs $2^4 - 1 = 15$ probabilities. The chain rule rewrites it but does not shrink it:

$$p(T, J, R, S) = p(T \mid J, R, S)p(J \mid R, S)p(R \mid S)p(S)$$

Now bring in **background knowledge** — assumptions of conditional independence:

Tracey's grass does not depend on Jack's

$$p(T \mid J, R, S) = p(T \mid R, S)$$

Jack's grass does not depend on Tracey's sprinkler

$$p(J \mid R, S) = p(J \mid R)$$

Rain is not influenced by the sprinkler

$$p(R \mid S) = p(R)$$

$$p(T, J, R, S) = p(T \mid R, S)p(J \mid R)p(R)p(S)$$

Only $4 + 2 + 1 + 1 = 8$ probabilities remain — and each one is a question a person can actually answer.

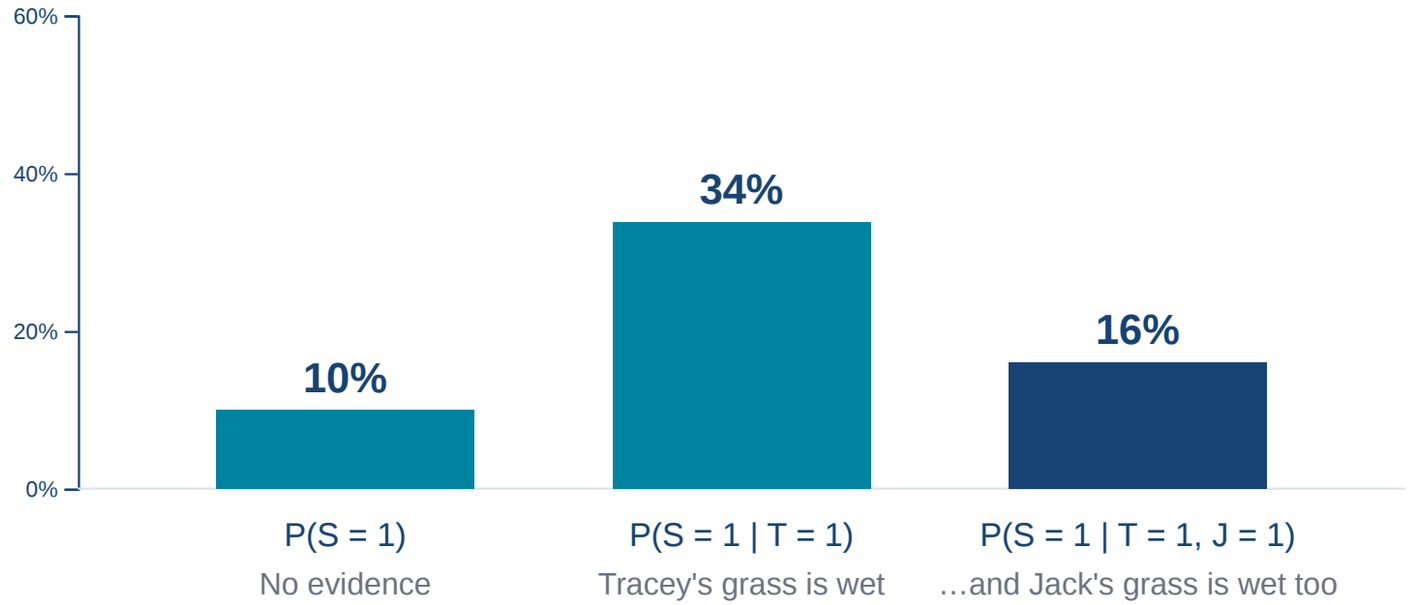
The Probability Tables

Factor	$p(\cdot)$	$1 - p(\cdot)$
$p(R = 1)$	0.2	0.8
$p(S = 1)$	0.1	0.9
$p(J = 1 \mid R = 0)$	0.2	0.8
$p(J = 1 \mid R = 1)$	1.0	0.0
$p(T = 1 \mid R = 1, S = 0)$	1.0	0.0
$p(T = 1 \mid R = 1, S = 1)$	1.0	0.0
$p(T = 1 \mid R = 0, S = 1)$	0.9	0.1
$p(T = 1 \mid R = 0, S = 0)$	0.0	1.0

Missing edges in the graph are exactly the assumptions above; the tables supply the numbers.

Inference: What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



Explaining Away

$$p(S=1 \mid T=1, J=1) = \frac{\sum_R p(T{=}1 \mid R, S{=}1)p(J{=}1 \mid R)p(R)p(S{=}1)}{\sum_{R,S} p(T{=}1 \mid R, S)p(J{=}1 \mid R)p(R)p(S)} = \frac{0.0344}{0.2144} \approx 0.16$$

Two independent causes (R and S) become **dependent once their common effect is observed**. Evidence for rain — Jack's wet grass — lowers the probability of the sprinkler from 0.34 to 0.16, even though rain and sprinkler are unconditionally independent.

Takeaway

A graphical model turns **background knowledge into a factorisation**: missing edges are conditional independence assumptions, and each one removes parameters. The graph is a picture of how the joint splits — nothing more, and nothing less.

Module 3

Bayesian Belief Networks

> ***Guiding Question:*** *What exactly does an arrow in a probabilistic model claim — and what does a missing arrow claim?*

Definition

Bayesian belief network (BBN). A **directed acyclic graph** whose nodes are random variables, in which the joint distribution factorises as

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p\big(x_i \mid \mathrm{pa}(x_i)\big)$$

where $\mathrm{pa}(x_i)$ are the parents of x_i in the graph.

Each node carries one conditional probability table given its parents. The wet-grass model was exactly this.

What the graph claims: each variable is independent of its non-descendants given its parents. A *missing* arrow is an independence claim; an arrow only permits a dependence.

Three Tasks

Modelling

Choose variables, edges and tables that encode the domain.

Inference

Compute posteriors given evidence — at a cost set by the graph's shape.

Learning

Estimate the tables — and possibly the structure — from data.

Modelling Technique: Divorcing Parents

A node with k binary parents needs 2^k table entries. **Divorcing** inserts an intermediate variable that summarises a group of parents, so the table becomes two small ones instead of one large one.

The intermediate node usually has a meaning of its own — "total load", "combined risk" — which is why the technique improves interpretability as well as size.

> Fewer parameters means fewer numbers a domain expert has to invent.

Modelling Technique: Noisy OR

When many causes $x_1, \ldots, x_k$ can each independently trigger an effect y, a **noisy OR** models the effect as "each cause fires with its own probability, and the effect occurs if any of them fires":

$$p(y = 0 \mid x_1, \ldots, x_k) = \prod_{i:x_i=1} (1 - \lambda_i)$$

k parameters instead of 2^k.

Typical use

Disease–symptom networks, where many diseases give rise to the same symptom.

Uncertain and Unreliable Evidence

Uncertain (soft) evidence

The state of x is itself uncertain, expressed as a distribution $\tilde{x}$. Infer by summing over the states:

$$p(y \mid \tilde{x}) = \sum_{x} p(y \mid x) p(x \mid \tilde{x})$$

We look at the grass without touching it: $p(T{=}1 \mid \tilde{T}) = 0.8$.

Unreliable evidence

The report is kept, but its table is **replaced** by our own judgement of how reliable it is:

$$p(x \mid \mathrm{pa}(x)) \to \tilde{p}(x \mid \mathrm{pa}(x))$$

Jack says his grass is wet, but Jack is in a bad mood and did not check.

Both are drawn with a **dashed** edge.

Takeaway

A belief network is a **DAG plus one conditional table per node**. Modelling techniques — divorcing, noisy OR, soft evidence — all trade assumptions for parameters, and every trade should be stated out loud.

Three Ways Three Nodes Can Connect

Every conditional-independence question in a DAG reduces to these three patterns.

Chain

$$a \to c \to b$$

a and b are dependent — but **independent given** c.

Information flows along the chain until the middle is observed.

Common cause (fork)

$$a \leftarrow c \to b$$

a and b are dependent — but **independent given** c.

The shared cause explains the correlation.

Common effect (collider)

$$a \to c \leftarrow b$$

a and b are **independent** — but **dependent given** c.

Explaining away — the reverse of the other two.

Mini-Exercise

Rain and the sprinkler are independent: rain → wet grass ← sprinkler. Your grass is **wet**, and you learn that it **rained**.

Does that change how likely it is that the sprinkler was on?

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Does that change how likely it is that the sprinkler was on?

Yes — it drops. Rain already explains the wet grass. Observing the common effect made two independent causes dependent: *explaining away*, the collider case. On a chain or a fork, observing the middle does the opposite.

d-Separation

A path between X and Y is **blocked** by a set Z if it contains either

- a chain or fork whose middle node **is in** Z, or
- a collider whose middle node — and all of its descendants — are **not in** Z.

If every path from X to Y is blocked by Z, then X and Y are **d-separated** given Z, and $X \perp\!\!\!\perp Y \mid Z$.

d-separation is sound: what it declares independent really is independent for every distribution factorising over the graph. It is not complete for a *particular* distribution — extra independences may hold numerically without being visible in the graph.

Takeaway

Independence in a belief network is a **graph property**: chains and forks are blocked *by* observing the middle node, colliders are opened by it. d-separation is that rule applied to every path.

The Churn Shop, Drawn

App users churn less (Lecture 02: the gap is real). Two graphs explain it:

FORK

app ← loyalty → churn

Loyal customers install the app *and* stay. The graph claims $\text{app} \perp\!\!\!\perp \text{churn} \mid \text{loyalty}$.

DIRECT EDGE

app → churn ← loyalty

The app itself keeps customers. App and churn stay dependent **given** loyalty.

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The graphs disagree on one checkable claim. Compare app users and non-users **of equal loyalty** (say, the same tenure): equal churn favours the fork, a remaining gap the direct edge.

Module 4

Markov Networks

> ***Guiding Question:*** *Pixels in an image influence each other symmetrically — no pixel causes its neighbour. How do we model that?*

Undirected Models

Markov network (Markov random field). An **undirected** graph in which the joint distribution factorises over the **cliques** $c \in \mathcal{C}$ of the graph:

$$p(x_1, \ldots, x_n) = \frac{1}{Z} \prod_{c \in \mathcal{C}} \phi_c(x_c), \qquad Z = \sum_x \prod_c \phi_c(x_c)$$

The $\phi_c \geq 0$ are **potentials** — arbitrary non-negative functions, not probabilities. Z is the normalising **partition function**.

When all cliques have size 2 the model is a **pairwise Markov network**.

Directed vs. Undirected

Belief network

Directed, acyclic. Factors are conditional probabilities. Natural for **causal or generative** stories; conditioning on a collider creates dependence.

Markov network

Undirected. Factors are potentials with a global Z. Natural for **symmetric interactions** — pixels, atoms, neighbours in a lattice.

Marginalising and Conditioning Behave Oppositely

Marginalising over C

makes A and B **dependent** — the sum over all states of c couples the potentials.

Conditioning on C

In the chain $A - C - B$, a fixed c is a constant in the potentials, so the factor splits and $A \perp\!\!\!\perp B \mid C$.

This holds because C separates them — not for every C.

Markov Properties and the Separation Test

For undirected models, independence is plain graph separation: if every path from X to Y passes through Z, then $X \perp\!\!\!\perp Y \mid Z$.

For a **directed** model the same question is answered by converting the graph:

1. Ancestral graph

Keep $X \cup Y \cup Z$ and their ancestors; delete all other nodes and their edges.

2. Moralisation

Connect any two nodes that share a child, then drop all arrow heads.

3. Separation

Remove the nodes in Z and their edges. If no path joins X to Y, then $X \perp\!\!\!\perp Y \mid Z$.

Two Examples

Boltzmann machine

A pairwise Markov network over binary units with weights on the edges; the restricted variant (RBM) allows edges only between a visible and a hidden layer.

Image cleanup

Pixels form a lattice: each potential prefers a pixel to agree with its neighbours and with the noisy observation. Inference denoises the image.

Other Graphical Models

Chain graphs

Combine directed and undirected edges in one model.

Factor graphs

Make each factor an explicit node — mostly used **for inference**, since they state precisely which variables each factor touches.

Undirected models trade causal direction for symmetric interaction, and local normalisation for a global partition function. Separation in the (moralised) graph is the single test for conditional independence.

Module 5

Learning and Inference

> ***Guiding Question:*** *We can now write the model down. Can we actually compute anything with it?*

Three Questions You Can Ask a Model

Infer unobserved variables

Compute the posterior given evidence — probabilistic inference.

Learn the parameters

Estimate the factors' distributions from data.

Learn the structure

Discover which dependencies exist at all.

With complete data, learning the parameters is the easiest of the three — it is counting. Learning the structure is the hardest.

Inference Is Hard in General

Exact inference in belief networks is **NP-hard**, and so is approximate inference with a guaranteed error bound. Whether a particular model is tractable depends on its structure.

Exact inference

Variable elimination, junction trees, message passing.
Efficient when the graph is a tree or has low treewidth.

Approximate inference

Sampling (MCMC, importance sampling) and variational methods.
Trade the guarantee for tractability.

Learning the Parameters

With **fully observed** data, parameter learning is counting: each conditional table is estimated from the frequencies of its variable and parents.

With **hidden variables**, the counts are unavailable and one alternates between inferring the hidden values and re-estimating the parameters (expectation–maximisation).

> The estimates are only as good as the sample supporting each table row. A parent combination that never occurs in the data leaves its row undetermined — which is another argument for keeping tables small.

Takeaway

> Graphical models make the joint distribution **writable**; whether they make it **queryable** depends on the graph's structure. Parameters come from counting where data is complete, and from iterating where it is not.